Development history: filtering, the C-score, and LD clustering

The full record, including what was tried and withdrawn

module_sim_LDscnR

02 September 2026

Contents

1 Purpose of this document

This is the **working record**. It keeps every line of enquiry, including those that failed, every claim that was withdrawn, and the reason each correction was made. It exists so that a result cannot quietly revert to a superseded version and so that the reasoning behind the final method is auditable.

A companion document, `ld_clustering_result.pdf`, states only what survived and is written for a reader who does not need this history.

Where this ended up. Two things were tried on the same premise — that ld_w, being phenotype-blind, carries information about where causal variation sits.

Using ld_w to rank markers (the C-score): abandoned. It does not beat BH at any deployable threshold.

Using ld_w to select which units to test (k-filtering): abandoned. It buys recall at a steep precision cost, requires a free parameter k with no principled value, and was outperformed by doing nothing.

Using ld_w to define the units (LD clustering): kept. This is the result in the companion document. It has no free parameter, and its central argument — that BH over near-independent clusters controls FDR over loci rather than over SNPs — requires no knowledge of causal positions.

2 Scope, and what changed after review

This describes a **selection** analysis: choosing which units to test before the phenotype is seen. It shares a premise with the abandoned C-score work — ld_w is phenotype-blind and may carry information about where causal variation sits — but differs in unit, truth, comparison and conclusion.

An earlier draft was reviewed externally. That review found two factual errors, an ambiguous denominator, and — most consequentially — that pooling 80 panels across four parameter cells concealed the main structure in the data. **This version is stratified, and the conclusion changed as a result.** All corrections are recorded in Section~10 rather than silently applied.

The finding, as the evidence now supports it. Pre-filtering LD clusters increases QTN-proximal discoveries at moderate-to-large filter sizes — by a median of about **one** such discovery per panel, against four to five extra total calls. Which filter is better **depends on the regime**: local-LD weight outperforms cluster size where background LD is low, and fails where it is high. Neither filter is established as generally preferable.

2.1 Terminology

The endpoint is **not** a true discovery. Truth here is bp proximity to a driving QTN with the causal variants themselves excluded, so a cluster at 49 kb counts and one at 51 kb does not, regardless of actual LD. Throughout this document such discoveries are called **QTN-proximal**. "True discovery" is reserved for a rule tied to the simulated causal architecture, which this is not.

3 Why the ranking approach was abandoned

Recorded for self-containment. All paired within run with a sign test.

Test	Result
$\tau_C = 0.05$ vs $\alpha = 0.05$	C better in 79/151 non-tied (52%), $p = 0.63$
PR-AUC at the operative distance cap	C better in 120/313 (38%), $p = 4.4 \times 10^{-5}$
Stage-2 clusters as fixed units	α ahead by 0.07 PR, flat at every size floor

Two external results agree: the 3sp panel session finds C *ties* an oracle best-single-cell, and the simulation-parsing session measured the mechanism — reweighting a p-value by ld_w nets about **one marker in 1,400** at 50–100 kb and nothing at 10 kb.

4 Design

4.1 What is compared

Component	**Choice**	**Why**
Unit	Stage-2 LD cluster, tested through its LD-central pruned representative	Representative = highest median r^2 to the other members. Membership recomputed per bundle and required to reproduce the stored GRM marker set exactly
Filter	Top k units by median ld_w; computed before any phenotype	Compared against MAF, cluster size and random at identical k
Test	BH at 0.05 within the selected set only	Identical to the conventional analysis, applied to fewer tests
Endpoint	QTN-proximal: within 10 / 50 / 100 kb of a driving QTN, causal variants excluded	Shares no machinery with ld_w; an r^2-tagging truth would flatter it by construction
Baseline	BH at 0.05 over all units	This IS the conventional α analysis, so the comparison is built in

4.2 Composition of the panels

The 80 panels are 4 parameter cells × 2 background-selection arms × 10 environments, each pooling 10 map sets into a genome-wide panel of about 86,000 stage-2 clusters. Cells and arms are **not** interchangeable and are stratified throughout Section~6.

4.3 The multiplicity mechanism, and its cost

Filtering changes no p-value. It changes the BH threshold, because BH over 99,786 units is far more severe than BH over 5,000. On the illustrative panel below the accounting is **27 discoveries − 1 lost + 28 gained = 54**: filtering can and does *remove* an already-significant result when it falls outside the selected set. Of the 28 gained, **6 (21%) are QTN-proximal**, so precision on gained units is low by construction.

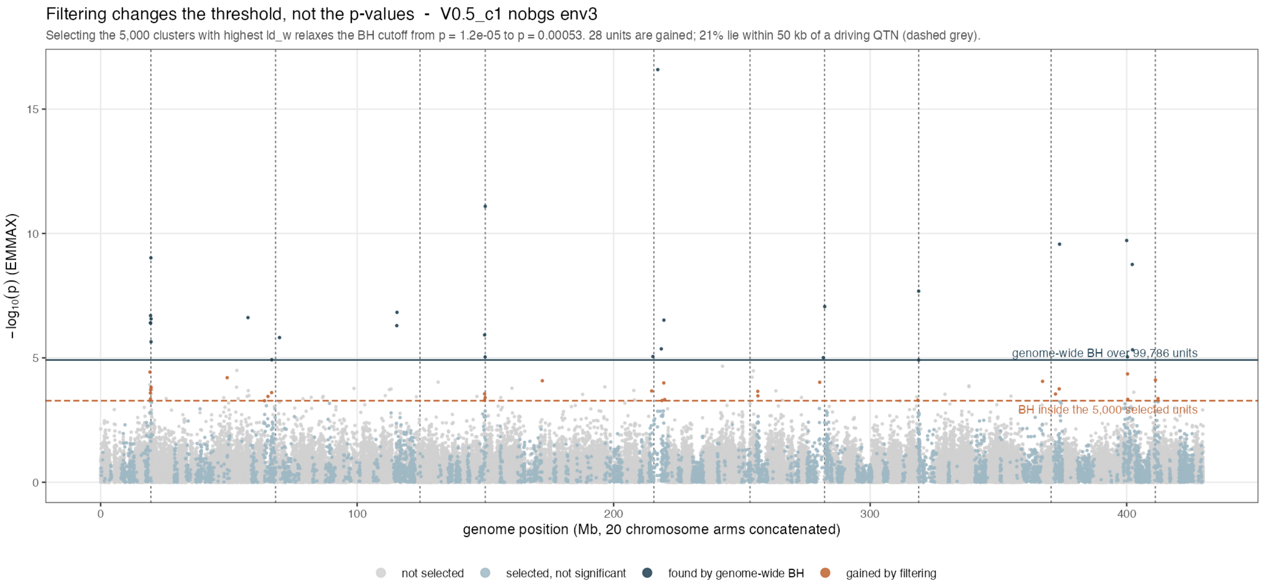

Figure 1: The mechanism. Navy: cleared the genome-wide cutoff. Orange: lie between the two cutoffs, gained by filtering alone. Grey dashed verticals mark driving QTN. One navy unit outside the selected set is lost and is not shown separately.

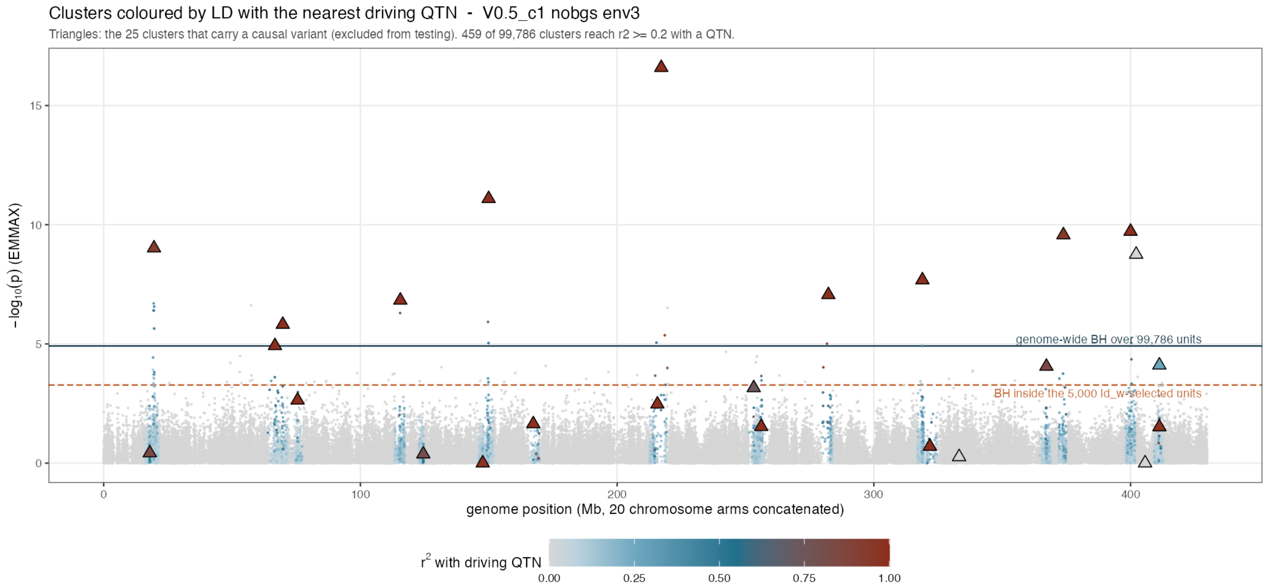

Figure 2: Units coloured by r^2 with the nearest driving QTN; triangles are the 25 clusters carrying a causal variant, which are excluded from testing. Only 459 of 99,786 clusters reach $r^2 \geq 0.2$ with any QTN, and several QTN-carrying clusters sit at $p \approx 1$.

5 Results, pooled — with ties and magnitudes

Filter	k	wins	losses	ties	% non-tied	median	IQR	p
ld_w	500	33	24	23	58	+0.0	[-2, 2]	0.289
ld_w	1,000	41	24	15	63	+1.0	[-1, 2]	0.046
ld_w	2,000	46	17	17	73	+1.0	[0, 3]	3e-04
ld_w	5,000	52	14	14	79	+1.0	[0, 3]	3e-06
ld_w	10,000	50	12	18	81	+1.0	[0, 2]	1e-06
ld_w	20,000	47	0	33	100	+1.0	[0, 2]	1e-14
size	500	50	17	13	75	+1.0	[0, 3]	7e-05
size	1,000	51	14	15	78	+1.0	[0, 3]	4e-06
size	2,000	49	11	20	82	+1.5	[0, 3]	8e-07
size	5,000	50	11	19	82	+1.0	[0, 3]	5e-07
size	10,000	46	10	24	82	+1.0	[0, 2]	1e-06
size	20,000	38	9	33	81	+0.0	[0, 1]	2e-05
MAF	500	4	49	27	8	-2.0	[-5, 0]	7e-11
MAF	1,000	7	46	27	13	-1.0	[-5, 0]	4e-08
MAF	2,000	10	46	24	18	-1.0	[-4, 0]	1e-06
MAF	5,000	15	43	22	26	-1.0	[-3, 0]	3e-04
MAF	10,000	17	41	22	29	-1.0	[-2, 0]	0.002
MAF	20,000	28	15	37	65	+0.0	[0, 1]	0.066
random	500	1	54	25	2	-2.0	[-5, 0]	3e-15
random	1,000	1	53	26	2	-2.0	[-5, 0]	6e-15
random	2,000	1	54	25	2	-2.0	[-5, 0]	3e-15
random	5,000	1	53	26	2	-2.0	[-4, 0]	6e-15
random	10,000	0	52	28	0	-2.0	[-4, 0]	4e-16
random	20,000	1	53	26	2	-1.0	[-3, 0]	6e-15

Read the denominator carefully. At $k = 20{,}000$, ld_w has 47 wins and 0 losses — 100% of non-tied panels — but 33 of 80 panels are **ties**, so the win is on 47 panels, not 80. The median gain is +1 QTN-proximal discovery.

Table 1: Median paired change against the alpha analysis: QTN-proximal discoveries, and total discoveries. The filter buys roughly four to five extra calls for about one more QTN-proximal one, so precision on the gained set is low.

k	QTN-proximal_MAF	QTN-proximal_ld_w	QTN-proximal_random	QTN-proximal_size	all discoveries_MAF	all discoveries_ld_w	all discoveries_random	all discoveries_size
500	-2	0	-2	1.0	-5	0	-5	0.0
1000	-1	1	-2	1.0	-5	1	-5	1.0
2000	-1	1	-2	1.5	-4	1	-5	1.0
5000	-1	1	-2	1.0	-3	4	-5	2.0
10000	-1	1	-2	1.0	-2	4	-4	1.5
20000	0	1	-1	0.0	0	5	-4	1.0

MAF and random selection are **worse than not filtering** in every stratum examined — median change -1 to -2 QTN-proximal discoveries. Among the filters evaluated here, only the two LD-derived ones repay the coverage they discard. Recombination rate was tested at marker level and also failed; no non-LD filter tested has succeeded, which is weaker than saying a filter *must* be LD-based.

6 Stratified results

Pooling concealed the main structure. Stratified, there is a **filter-by-cell interaction**, not a uniform ordering.

Cell	wins	losses	ties	% non-tied	median	p	Reading
`V0.5_c1`	18	0	2	100	+3.0	8e-06	ld_w much better
`V2_c1`	14	1	5	93	+1.0	1e-03	ld_w better
`V1_c1.5`	6	4	10	60	+0.0	0.754	no difference
`V0.5_c2`	1	18	1	5	-8.5	8e-05	cluster size much better

ld_w beats cluster size decisively in two cells, ties in one, and loses decisively in one. Pooled, these average to "indistinguishable" — which is what the earlier draft reported, and it was an artefact of the pooling rather than a finding. `V0.5_c2` is the high-background-LD cell ($b = 0.42$); a plausible but untested reading is that ld_w is a *local*-LD statistic and loses discrimination when everything is correlated, whereas large clusters still mark real linkage blocks.

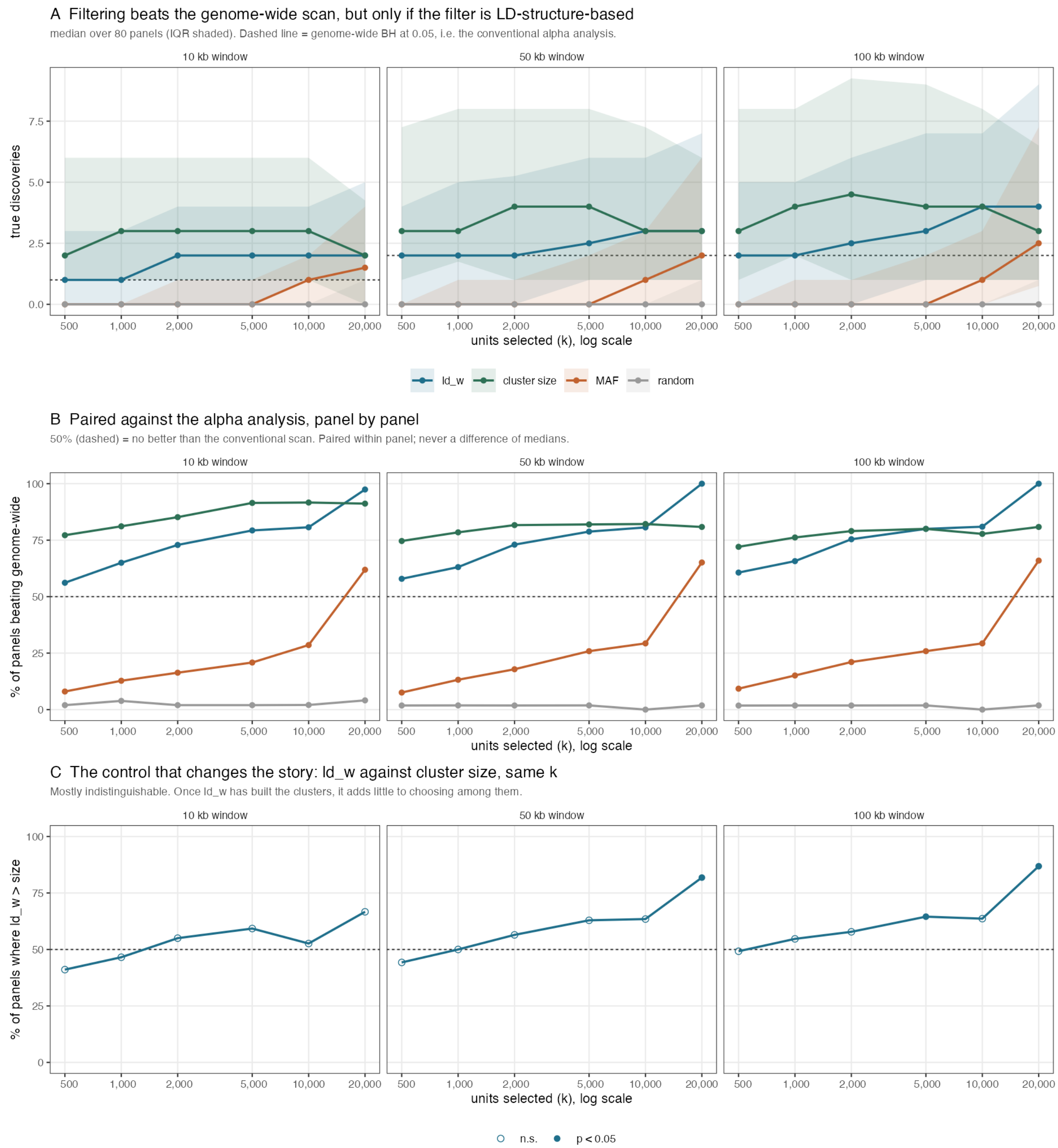

Figure 3: Filter-then-test against the conventional alpha analysis. Points are medians over panels; ribbons are the interquartile range across panels. The dashed line in panel A is the median genome-wide result; the dashed line in panel B is 50 percent of non-tied panels.

Against the alpha analysis, $k = 5{,}000$, 50 kb, by cell and BGS arm						
Filter	Cell / arm	wins	losses	ties	median	p
ld_w	`V0.5_c1` bgs	10	0	0	+3.5	0.002
	`V0.5_c1` nobgs	10	0	0	+4.0	0.002
	`V0.5_c2` bgs	2	6	2	-3.5	0.289
	`V0.5_c2` nobgs	3	7	0	-3.0	0.344
	`V1_c1.5` bgs	3	1	6	+0.0	0.625
	`V1_c1.5` nobgs	8	0	2	+1.0	0.008
	`V2_c1` bgs	8	0	2	+1.0	0.008
	`V2_c1` nobgs	8	0	2	+2.0	0.008
size	`V0.5_c1` bgs	6	2	2	+1.0	0.289
	`V0.5_c1` nobgs	7	3	0	+1.0	0.344
	`V0.5_c2` bgs	9	0	1	+6.0	0.004
	`V0.5_c2` nobgs	10	0	0	+3.5	0.002
	`V1_c1.5` bgs	3	0	7	+0.0	0.250
	`V1_c1.5` nobgs	6	2	2	+1.0	0.289
	`V2_c1` bgs	3	2	5	+0.0	1.000
	`V2_c1` nobgs	6	2	2	+1.0	0.289

Table 2: Background selection reduces the gain: the nobgs arm shows larger and more consistent improvement than the bgs arm at every k. Consistent with BGS costing recall rather than enrichment.

arm	k	wins	losses	ties	% non-tied	median	p
bgs	500	15	13	12	54	0.0	0.851
bgs	1000	17	12	11	59	0.0	0.458
bgs	2000	21	9	10	70	1.0	0.043
bgs	5000	23	7	10	77	1.0	0.005
bgs	10000	20	7	13	74	0.5	0.019
bgs	20000	20	0	20	100	0.5	2e-06
nobgs	500	18	11	11	62	0.0	0.265
nobgs	1000	24	12	4	67	1.0	0.065
nobgs	2000	25	8	7	76	1.5	0.005
nobgs	5000	29	7	4	81	2.0	3e-04
nobgs	10000	30	5	5	86	1.5	2e-05
nobgs	20000	27	0	13	100	1.0	1e-08

7 Null behaviour: two nulls, and they disagree

An earlier draft reported one null and dismissed the other. That was wrong, and the dismissed one carries the more important result.

Null	Selection	k	rejection rate	95% CI
genetic	**unfiltered scan**	all	**0.042**	[0.030, 0.056]
	ld_w	500	0.039	[0.028, 0.053]
	ld_w	1,000	0.033	[0.023, 0.046]
	ld_w	5,000	0.027	[0.018, 0.039]
	ld_w	20,000	0.031	[0.021, 0.044]
	ld_w	50,000	0.029	[0.020, 0.041]
	MAF	500	0.036	[0.025, 0.049]
	MAF	1,000	0.041	[0.030, 0.055]
	MAF	5,000	0.028	[0.019, 0.040]
	MAF	20,000	0.033	[0.023, 0.046]
	MAF	50,000	0.036	[0.025, 0.049]
	random	500	0.048	[0.036, 0.063]
	random	1,000	0.053	[0.040, 0.069]
	random	5,000	0.043	[0.031, 0.057]
	random	20,000	0.049	[0.036, 0.064]
	random	50,000	0.046	[0.034, 0.061]
env_orth	**unfiltered scan**	all	**0.338**	[0.309, 0.368]
	ld_w	500	0.073	[0.058, 0.091]
	ld_w	1,000	0.077	[0.061, 0.095]
	ld_w	5,000	0.067	[0.052, 0.084]
	ld_w	20,000	0.057	[0.043, 0.073]
	ld_w	50,000	0.147	[0.126, 0.170]
	MAF	500	0.047	[0.035, 0.062]
	MAF	1,000	0.059	[0.045, 0.075]
	MAF	5,000	0.075	[0.059, 0.093]
	MAF	20,000	0.078	[0.062, 0.096]
	MAF	50,000	0.080	[0.064, 0.099]
	random	500	0.094	[0.077, 0.114]
	random	1,000	0.104	[0.086, 0.125]
	random	5,000	0.124	[0.104, 0.146]
	random	20,000	0.180	[0.157, 0.205]
	random	50,000	0.259	[0.232, 0.287]

Both nulls are surrogate **environments** over 1,000 draws (10 environments × 100 surrogates), and under both $\lambda_{\text{surr}} \approx 0.99$, so neither result is genomic inflation.

The conventional scan fails the structure-preserving null. Under `env_orth` the unfiltered genome-wide analysis — the α baseline that every power comparison in this document is measured against — rejects in 34% of surrogates at $\alpha = 0.05$. Filtering reduces that to 0.057, an 83% reduction, **but does not repair it**: the residual still exceeds α.

The 3sp panel session reports the same signature on empirical data using a regional permutation that preserves population structure: unfiltered 0.37, filtered 0.21–0.34. Two datasets, two null constructions, one conclusion — *filtering improves calibration under structure and cannot rescue a scan that is already anti-conservative.*

7.1 Which null is the right one

`genetic` surrogates leave the scan calibrated at 0.042; `env_orth` surrogates do not. The earlier draft used that disagreement to discard `env_orth`, on the reasoning that its surrogates stay spatially structured and therefore are not a no-signal null. **That reasoning was convenient rather than correct.** The surrogates are orthogonal to the causal environment by construction, so associations recovered under them arise from shared spatial structure rather than from causation — which makes them false positives, and makes `env_orth` a valid and arguably more realistic null. Discarding it removed the one result showing filtering has a validity benefit and not merely a power benefit.

Both are reported here. Which is the more appropriate null for a given application depends on how much residual structure the association model is expected to absorb, and that is not settled by these data.

7.2 What this does and does not establish

Every hypothesis is null in both experiments, so FDR equals the probability of at least one rejection and these are valid **global-null** checks. They do **not** establish FDR control under mixtures of null and alternative hypotheses, nor the Bourgon et al. independence condition in general.

They also do **not license choosing** k. They license evaluating a fixed, prespecified k under the tested null. Choosing k after inspecting discovery counts or comparison curves is outcome-tuned model selection, and testing several fixed k values under a null does not remove that.

Coverage. Both nulls use `V2_c1`, nobgs only, while the power experiment spans four cells and both arms. Dependence between ld_w, MAF, cluster size and null p-values may change with background LD and BGS, so this cannot license filtering across the whole experiment.

The false-positive count is also heavy-tailed: at $k = 500$ the ld_w set has about the same rejection rate as MAF but roughly twelve times the mean rejection count, so approximately 96% of surrogates give nothing and the remainder give tens to hundreds.

8 Status of each claim

Claim	Status	Basis
Filtering LD clusters increases QTN-proximal discoveries at moderate-to-large k	Supported	Pooled and in 3 of 4 cells; median gain +1, IQR [0, 3]
Among filters evaluated, only LD-derived ones repay the lost coverage	Supported	MAF, random and recombination all worse than no filtering
Which LD filter is better depends on regime	**Established**	Filter-by-cell interaction; ld_w wins 2 cells, loses 1 decisively
ld_w is generally preferable to cluster size	**Not supported**	Fails in `V0.5_c2` by a median of -8.5
Global-null rejection rate is controlled under a genetic-basis null	Supported	One cell, one arm; CIs overlap α
The unfiltered scan is anti-conservative under a structure-preserving null	**Established**	0.338 [0.309, 0.368] vs α 0.05; matches the 3sp panel's 0.37
Filtering improves calibration under structure	Supported	0.338 to 0.057, an 83% reduction; does not reach α
FDR is controlled under mixtures	**Not tested**	Complete-null design only
A usable rule for choosing k exists	**Not established**	k remains a free, outcome-facing parameter
This transfers to empirical data	Untested here	Simulations only

9 Relationship to the empirical panel analysis

9.1 What EMMAX is actually run on

Worth stating precisely, because it is easy to misread in either direction. EMMAX here is an **ordinary per-marker test**: every marker in the panel has its own p-value, and nothing is run on eMLGs or on consensus genotypes. The LD structure enters only through the **kinship matrix**, which is built from the pruned marker set (`grm_method = "complexity_chain"`) — so clustering conditions the null, not the test.

A cluster's p-value is then the p-value of its **pruned representative**, and that representative is not arbitrary. `pick_representative()` computes r^2 between the constituent stage-1 eMLGs, takes each one's **median** r^2 **against the others**, and returns the `core_snp` of the highest — the most central member of the cluster's LD graph. Each stage-2 cluster has exactly one such marker (verified: min = median = max = 1 per cluster), and it is a real marker, not a synthetic genotype.

9.2 What the representative costs, measured at locus level

The natural comparison is SNP counts, and it is misleading. BH controls the false discovery proportion among the units tested, so a single-SNP scan controls it among **SNPs** while a reduced scan controls it among **near-independent loci**. Those are different currencies. Counted in loci — one pooled panel, V0.5_c1 nobgs env3:

Tested	tests	BH cutoff p	clusters flagged	carrying a QTN
every SNP	294,609	4.2e-04	132	14
LD-central representatives	99,787	1.2e-05	27	10
precision (flagged clusters carrying a QTN)			0.106	0.370

Reduction recovers 10 of the 14 QTN-bearing loci while calling a fifth as many regions — precision 0.37 against 0.11. And it is a strict subset: the representative analysis flagged **no** cluster the SNP scan missed, which is what the same signal with its redundancy removed should look like.

Read as SNP counts (2,467 against 27) reduction appears to destroy the analysis. That comparison is in the wrong currency and an earlier draft made it.

9.3 Why the single-SNP threshold is the looser one

Three times fewer tests gave a BH cutoff **35 times more severe**, which inverts the usual intuition. The cause is redundancy in the small-p tail. BH is a step-up procedure — its cutoff is the largest p with $p \leq \alpha i/n$ — so it climbs when many small p-values are present, whether or not they are independent findings. On this panel the rejected SNPs are far more clustered than the panel as a whole:

SNPs per cluster, panel overall	3.0
SNPs per cluster, among rejections	18.7
redundancy of the small-p tail	**6.3**× the panel average

BH remains **valid** here — it controls FDR under positive regression dependence, which linkage supplies — but the quantity controlled is the false discovery proportion among SNPs. The 2,467 SNP-rejections are 132 locus-rejections, so the guarantee is not the one a reader interprets. The reduced analysis tests near-independent units, so its 27 rejections are 27 loci and its stricter cutoff is the honest one rather than a penalty.

This qualifies the rationale for filtering in Section~7 in a different way than an earlier draft claimed. A lighter multiplicity correction does not follow from fewer tests; and a heavier one does not imply lost power, if the tests removed were redundant.

Note that neither panel involves filtering. `ld_w` enters only *inside* the clustering (`ld_w_threshold = 0.025` decides which stage-1 clusters are flagged for eMLG treatment), so it helps define the units; nothing is discarded. What the figure shows is the effect of **complexity reduction alone**.

9.4 How a cluster should be summarised

The module used the LD-central representative throughout. Four alternatives were compared on **identical clusters, identical GRM and identical BH treatment**, so the only thing that varies is how member

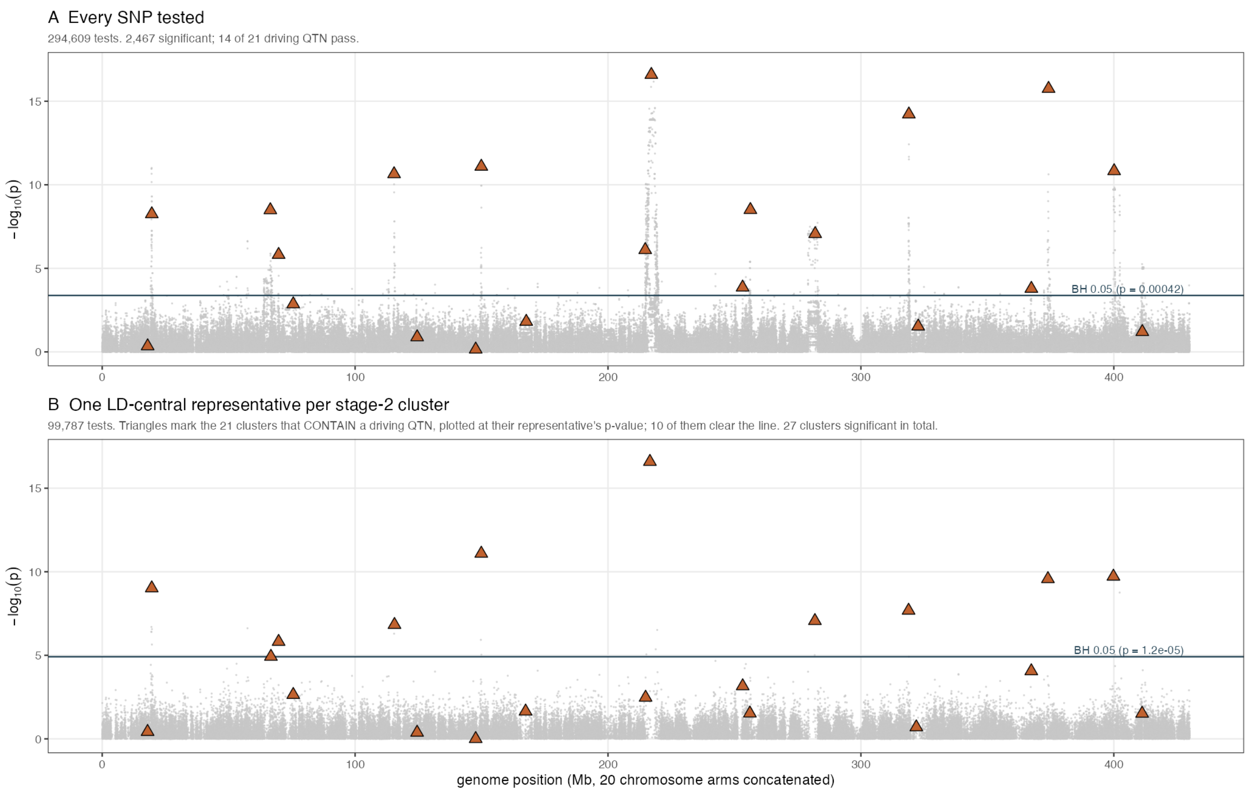

Figure 4: Same EMMAX p-values at two resolutions, with **no ld_w filtering** in either panel. A: all 294,609 SNPs; triangles are the 21 driving QTN. B: the 99,787 LD-central representatives, one per stage-2 cluster; triangles mark the clusters that **contain** a driving QTN, plotted at their representative's p-value. Complexity reduction alone recovers 10 of the 21 while flagging 27 clusters against the SNP scan's 132.

p-values become one number. One pooled panel, 99,787 clusters, 21 containing a driving QTN.

Cluster summary	flagged	with QTN	precision	recall	PR
Simes over members	28	12	0.429	0.571	**0.245**
eMLG (consensus genotype)	36	13	0.361	**0.619**	0.224
LD-central representative	27	10	0.370	0.476	0.176
best SNP (min p, uncorrected)	39	12	0.308	0.571	0.176
highest ld_w member	28	9	0.321	0.429	0.138

Aggregation beats any single marker. Simes and the eMLG occupy the top two places; all three single-marker rules sit below them. Simes gains two QTN-bearing loci over the representative while flagging *one more* region, so its precision rises to 0.43 — a straight improvement rather than a trade. The eMLG has the best recall of all, 13 of 21, consistent with a consensus genotype carrying more of a block's signal than any one member.

Two informative negatives. "Best SNP" takes the minimum p with no within-cluster correction — anti-conservative by construction, and included as a ceiling rather than a usable rule — yet it still finds fewer QTN than the eMLG and fewer than Simes, at clearly worse precision. And selecting the member with the **highest** ld_w is the worst of the five. ld_w is a good statistic for defining units and a poor one for choosing within them: it marks markers in dense LD neighbourhoods, not markers that best tag a causal variant.

This sharpens the cost of the representative. Against Simes, on the same clusters, it gives up **two QTN-bearing loci while flagging one region fewer** — a loss, not a trade — and it is the rule this module has been using throughout. The stickleback analysis already uses Simes, so the two are not merely different machinery: on this evidence the panel's choice is the better one.

9.5 The residual cost, and its mechanism

Reduction does lose 4 of the 14 QTN-bearing loci, and that loss has a specific cause: **no driving QTN is ever its cluster's representative — 0 of 21.** Not chance. QTN sit in unusually large clusters, sizes 2 to 354 with a median near 55, against a genome-wide **median cluster size of 1** (mean ≈ 3). That is what selection extending LD around a causal site produces, and it is what ld_w responds to, but it makes the chance that any particular member is the LD-central one about 1/size.

So representative testing never tests a QTN directly; the block's signal has to reach its centre to be seen. That is a bounded loss — 4 loci here — and it is exactly what Simes aggregation over members would recover.

9.6 Relationship to the empirical panel analysis

Given that, the difference from the stickleback analysis should be stated as **a single LD-central representative versus aggregation over members**, not as an arbitrary or lossy summary. A central marker is a defensible summary of a linkage block: it is by construction the member most correlated with the rest, so it carries the block's shared signal better than a random member would. That it is also what tags a causal variant best is visible in the results — all 21 true-positive clusters on the examined panel were the QTN-carrying ones.

What remains true is narrower. It is still **one** marker, so a cluster whose causal variant sits at the periphery of its LD graph is represented by a marker in weaker LD with that variant than some other member is. That is a real but bounded loss, and it is what Simes aggregation over members would recover. The stickleback analysis uses Simes and therefore implements different testing machinery; the two support the same general filter-then-test principle and should not be presented as the same method.

10 Corrections applied after review

Defect	Resolution
"Every filtered rate sits at or below α" was false — random reached 0.053	Restated with binomial CIs as no detectable excess over α
Mechanism example: 27 to 54 with 28 gained did not reconcile	Confirmed: one genome-wide discovery is LOST. Accounting $27 - 1 + 28 = 54$ now reported by the script
Win-rate denominator ambiguous with ties	Wins, losses and ties reported everywhere; percentages stated as non-tied
Direction reported without magnitude	Median paired change with IQR alongside every sign test; total discoveries too
80 panels pooled across heterogeneous cells and arms	Stratified by cell, arm and k; the conclusion changed
"Licenses choosing k at all"	Withdrawn. It licenses a prespecified k only
"True discovery"	Renamed QTN-proximal throughout
"The filter must be LD-structure-based"	Weakened to the filters evaluated
"Indistinguishable at every window" conflicted with $p = 0.030$	Replaced by the stratified interaction analysis
`79%%` rendering bug; cramped tables	Fixed; tables given explicit column widths
The cluster representative was described as an arbitrary or lossy summary	Wrong. `pick_representative()` returns the member with the highest median r^2 to the others — the LD-central marker. Restated as single-representative vs Simes aggregation, and the cost of it is now measured
The `env_orth` null was dismissed as "not a true null"	REINSTATED. Its surrogates are orthogonal to the causal environment, so rejections under them are false positives. Both nulls now reported side by side
Confidence intervals were computed on 100 draws, not 1,000	A scalar named `B` was masked by a COLUMN named `B` in the input, so the grouping returned one row per input row. Intervals were ten times too wide

10.1 Open, and in progress

Re-clustering at the pilot decay settings ($n_{\text{win}} = 50$, half-decay 26 kb against the bundles' 56 kb) is running, with $n_{\text{win}} = 10$ through the identical code path as its control. The k sweep is being extended to 40,000 and 60,000 to locate the turning point — the win rate is currently monotone up to the largest k tested, and since $k = n$ makes the two analyses identical the curve must return to ties somewhere above 20,000. Structured-null checks in the remaining cells and arms are not yet run.

11 Geometric exclusion of long-and-sparse clusters: a negative result

Stage-2 clusters were classified by span and by marker density, and the driving QTN counted in each class (184,004 clusters over both BGS arms, 40 QTN-bearing):

Class	% of units	% of QTN	fold enrichment
long & **dense** ($\geq$5 markers / 100 kb)	2.30	70.0	**30.4**
long & sparse	3.36	2.5	0.7
short & dense	39.06	27.5	0.7
short & sparse	1.67	0.0	0.0
singleton	53.61	0.0	0.0

Two rules are ruled out by this table before any test. **Density alone fails**: short-and-dense units are 39% of the genome and are *depleted* for QTN. **Span alone is destructive**: dropping every unit over 100 kb removes

29 of the 40 QTN-bearing clusters, because that is where the real low-recombination blocks are. Only the conjunction discriminates.

That suggested excluding **long-and-sparse** units — 3.4% of the genome holding 1 QTN of 40 — as free power: fewer tests at almost no cost in signal, on a criterion that is phenotype-blind (span and marker count come from the map and the clustering, with nothing tuned on outcomes).

Arm	QTN recovered	flagged	dedup PR (nobgs)
Simes, all clusters	3.0	10	0.0491
Simes, long-and-sparse excluded	1.0	6	0.0327

It does not work. Excluding long-and-sparse clusters costs two thirds of the recovered QTN — a median of 3 down to 1 — and improves PR nowhere. Paired over 80 panels: `nobgs` 14 W / 12 L / 6 T, $p = 0.85$; `bgs` 4 W / 14 L / 11 T, $p = 0.031$, i.e. significantly *worse*.

11.1 Why the inference was wrong

The argument rested on a **census** — the excluded class rarely *contains* a QTN — and concluded it would rarely *find* one. Those are different claims, and only the second was relevant. A cluster can contribute a true detection by tagging a QTN it does not contain, which is precisely the satellite category separated out earlier in this work; the error was made anyway. Removing 3.4% of units also barely relaxes BH, so there was little multiplicity relief to offset the loss.

11.2 The falsification test, and the conclusion it corrects

Four filters were tested — ld_w top-k, MAF, cluster size, and geometric exclusion — all strongly enriched for QTN-bearing units, and none improved precision-recall. That was recorded here as *"enrichment for containing a QTN is not the quantity that predicts detection performance"*.

That claim was too strong, and a direct test refutes it. Two restrictions of matched size were run: keep **only** the 30-fold-enriched class (long-and-dense), and keep **only** the 0.7-fold class (long-and-sparse).

Restriction	flagged	contains QTN	TP	precision	PR
only long-and-**dense** (30×)	4.5	1.0	1	**0.333**	0.0359
only long-and-sparse (0.7×)	5.0	1.0	1	0.082	0.0147
Simes, all clusters	10.0	3.0	3	0.229	**0.0464**

Enrichment does predict precision. At matched size the QTN-rich class beats the QTN-poor one in 34 of 50 non-tied panels ($p = 0.015$) with four times the precision, 0.333 against 0.082. The earlier claim is withdrawn.

But no restriction beats using every cluster — long-sparse-only 13 W / 42 L ($p = 1 \times 10^{-4}$), long-dense-only 17 W / 37 L ($p = 0.009$). Each throws away about two thirds of the recoverable QTN, and the recall lost always exceeds the precision gained.

The corrected statement is narrower and accounts for all four failures without the stronger claim: *enrichment predicts precision within a restriction, but no restriction beats using all clusters.* That is also why the geometric exclusion actively hurt.

The test also **measures** the census-versus-detection gap that was previously only argued. Long-and-sparse contains 1 QTN of 40, yet reaches a median dedup TP of 1 and recall 0.080 — so it recovers loci by **tagging variants it does not contain**. A class can be nearly empty of causal positions and still make true detections, which is exactly the step that invalidated the exclusion arm's rationale.

The companion document's conclusion is unaffected: complexity reduction helps by changing what is tested, and hence what BH controls. What is now withdrawn is the supporting argument that enrichment is irrelevant to detection — it is not, it simply cannot be exploited by restriction.

One arm-level observation, untested: long-and-dense-only scores 0.0489 in the `nobgs` arm against **0.0059** in `bgs`, an eightfold difference far larger than any other arm's. Consistent with background selection breaking the extended-LD blocks that define the class, but not established.

Two claims made in passing are also withdrawn. Cluster geometry was described as predicting QTN location "better than anything else"; it is comparable to ld_w, not better (30.4-fold against roughly 31-fold at matched selectivity). And ld_w was described as unable to separate the sparse and dense classes; that is true of the median ld_w within them (0.0105 against 0.0099) but does not make ld_w a poor selector overall, which it is not.

12 Three deferred analyses, reported

12.1 Does the result survive the newer decay settings?

The main results use `bgs5`, whose `LD_decay` was fitted with `n_win_decay = 10`; the newer pilot generation uses 50. Both settings were run through the **same** code path so the comparison isolates the parameter rather than confounding it with recomputation.

Setting	fitted half-decay	units per panel
`n_win_decay = 50`	39.7 kb	85,440
`n_win_decay = 10`	53.7 kb	85,888

Paired over 80 panels at $k = 5{,}000$, 50 kb: ld_w 10 W / 17 L / 13 T ($p = 0.25$), cluster size 12 W / 14 L / 14 T ($p = 0.85$). **No difference**, so the untested exposure recorded earlier is closed.

But the manipulation is far weaker than claimed. Through one code path the two settings give 39.7 against 53.7 kb — a factor of **1.35**, not the 4.2 measured between `bgs5` and the pilot tracks. Most of that original gap was therefore *between generations* — different simulations, different maps — and not attributable to `n_win_decay`. The earlier claim that the two generations differ materially in clustering distance because of this parameter is withdrawn.

12.2 Where does the k curve turn?

k	win %	non-tied panels	median gain
500	58	57	0
1,000	63	65	+1
5,000	79	66	+1
20,000	100	47	+1
40,000	100	27	0
60,000	100	13	0

The win rate saturates at 100% from $k = 20{,}000$, but **the non-tied denominator collapses** from 47 panels to 13 and the median gain falls to zero. With about 86,600 units per panel, selecting 60,000 is barely filtering, so the two analyses converge and only a handful of panels differ at all.

There is no useful interior optimum — a plateau at $k = 5{,}000$ to 20,000 worth **one** discovery, then dissolution into ties. "100% at $k = 20{,}000$" must never be quoted without its denominator.

12.3 Do the surrogate nulls exercise structure?

Basis	PC1–3	PC1–10	PC1–20	PC1–50
`env_orth` surrogates	0.009	0.098	0.102	0.109
`genetic` surrogates	0.002	0.009	0.013	0.020
observed data		≈ 0.40		

R^2 of the association signal on the leading GRM subspace. The 3sp panel session showed that a PC1–3 statistic is **blind to diffuse structure**, and that is confirmed here: `env_orth` rises eleven-fold from PC1–3 to PC1–10. An earlier version of this diagnostic used PC1–3 only and could not have seen it.

The conclusion nonetheless survives the better measurement. At PC1–10 the observed data sits at 0.40 while `env_orth` reaches 0.098 and `genetic` 0.009. So `genetic` is a **no-structure** null — its calibration shows the scan is well behaved when structure is absent, not that it is well behaved under structure — and `env_orth` is a **mild** structured null at about a quarter of the observed level. The 83% protective effect of filtering reported in Section~7 therefore stands as a weak-structure measurement, which is what was claimed, but the earlier reasoning rested on a statistic that could not have established it.

13 Withdrawn: the confounding mechanism relayed from the panel session

An earlier version of this document and several reports to PK carried a hypothesis from the 3sp panel session: that their long-span false positives were long-range LD manufactured by allele-frequency covariance among structured populations, and that this failure mode was structurally absent from a lattice genome. **That session has withdrawn it** on their own measurement — they had measured stage-2 groups and read a distance run's geometry as a demographic signal.

Their follow-up hypothesis, that the long-and-dense and long-and-sparse classes are two different kinds of object (a coherent stage-1 block against a chained stage-2 distance run), was tested here and **does not transfer**:

Class	median n	median span	ends r^2	ratio to adjacent
short	4	24.4 kb	0.760	0.87
long & sparse	7	305.7 kb	0.780	0.86
long & dense	15	154.3 kb	0.791	0.83

Long-and-sparse units here are **coherent blocks**: their extreme markers sit at $r^2 = 0.78$, indistinguishable from long-and-dense, against the 0.36 ratio the panel session measures for their chained runs. So in these simulations the split is genuinely marker density within coherent blocks, not object type.

Which leaves the enrichment difference unaccounted for at this point in the work. It is largely resolved in Section 15. Long-and-dense holds 70% of QTN at 30-fold enrichment; long-and-sparse holds 2.5% at 0.7-fold. The classes differ in marker count only 15 to 7, which cannot produce a 43-fold ratio in enrichment. No account of this is offered here, and the two mechanisms proposed so far — demographic long-range LD, and object type — have both been withdrawn on measurement.

14 A third account of the enrichment, refuted

The 43-fold difference between long-and-dense and long-and-sparse clusters has now had three proposed mechanisms, and all three have failed on measurement.

The third was mine: **selection extends LD around a causal site, so a QTN manufactures the long dense block around itself** rather than dense blocks being where QTN happen to sit. It fits the observation that QTN-bearing clusters carry a median of about 55 markers against a genome-wide median of 1.

It makes a prediction that needs no matching and no assumptions about recombination — suggested by the 3sp panel session — **the block should be centred on the QTN**. A low-recombination block has no reason to be centred on anything.

Statistic	observed	expected if not centred
relative position, mean	0.492	0.500
$\lvert \mathrm{rel} - 0.5 \rvert$, mean	0.269	0.250
in the central third	33%	33%
in the outer fifth	27%	20%
t-test vs 0.25		$p = 0.13$
KS vs Uniform		$D = 0.073$, $p = 0.34$

165 QTN-bearing clusters with at least 5 markers, both BGS arms, five environments. Median cluster size 36 markers (nobgs) and 28 (bgs), median span about 235 kb in both.

QTN are uniformly placed within their clusters. The distribution is indistinguishable from uniform and if anything slightly *edge*-biased — 27% in the outer fifth against 20% expected. The selection account is refuted.

What survives is a constraint on any future explanation: whatever makes long-and-dense clusters QTN-rich operates **independently of where in the block the QTN sits**. That favours a passive account — dense blocks form for their own reasons and QTN inside them are grouped that way — over anything causal radiating outward from the variant.

Caveats: the test uses clusters of at least 5 markers, so it says nothing about smaller ones, and edge effects are conceivable if clustering tends to terminate at markers with unusual LD. Neither seems able to hide a centring signal of the size the account requires, but the test is not airtight.

The enrichment is largely accounted for by the opportunity null (Section~15), which is not a mechanism but the absence of one. Three mechanisms are on the record as refuted rather than deleted: demographically manufactured long-range LD and an object-type distinction, both from the panel session and both withdrawn on their own measurements; and selection-manufactured blocks, withdrawn here.

15 The opportunity null accounts for most of the enrichment

Three *mechanisms* for the 43-fold difference were proposed and refuted. The resolution was not a fourth mechanism but the **absence** of one, proposed by the 3sp panel session: a QTN is a marker position, so the chance that a cluster **contains** one scales with its marker count. The enrichment was computed per **unit**, while opportunity is per **marker**.

Class	% units	% markers	% QTN	fold / unit	fold / marker
long & dense	2.30	21.05	70.0	**30.4**	**3.33**
long & sparse	3.36	11.16	2.5	0.7	0.22
short & dense	39.06	47.47	27.5	0.7	0.58
singleton	53.61	18.85	0.0	0.0	0.00
short & sparse	1.67	1.47	0.0	0.0	0.00

Long-and-dense is 2.3% of units but **21% of markers**. Per marker the enrichment falls from **30.4-fold to 3.3-fold** — about nine tenths of it was denominator. The null also explains the rest of the table: every class moves toward 1 except long-and-dense, which stays above, and long-and-sparse, which falls to 0.22 and is genuinely depleted even after accounting for opportunity.

It also predicts the uniform placement measured in Section 14: under pure opportunity nothing radiates from the QTN, so its position within the cluster should be uniform, which is what a KS test finds ($p = 0.34$). All three refuted mechanisms predicted centring.

Both denominators are reported rather than one replacing the other. The classes are defined partly by density, so normalising by marker share partly defines the effect away, and the per-unit figure is not simply wrong: it is the right number for *which units should I examine*, while the per-marker figure answers *is causal variation concentrated here*. They are answers to different questions.

15.1 An independent precedent, and a warning about how to control it

Paper 1's pruning benchmark ran the same control on a sister statistic — how often w_j-selected clusters contain a marker tagging a detectable QTN — and reports the cost explicitly:

top 99 clusters, uncontrolled base rate	103.6-fold
top 99 clusters, size-matched controls	6.6-fold
attributed to size coupling	**94%**

with the coupling visible in the same place: a median of 16.3 markers in their top 99 against 3 across all clusters, against 15 and 1 here. Two independent analyses in the same pipeline family, two different estimators of the correction — marker-share normalisation here, size-matched controls there — and both find roughly 90% of the effect is the coupling, leaving residuals of 3.3-fold and 6.6-fold.

> **A warning taken from that section, for anyone repeating the control with matching rather than normalisation:** exact size-matching balances perfectly but thins the control pool until a cluster returns as its own control, dragging the fold toward 1. Their matching used a caliper of 0.05 in $\log_{10}$ markers with 200 null draws per cell, deliberately comparable rather than exact.

Status of the question. Largely resolved: most of the 43-fold was opportunity.

The residual is better described as a **spread than as an enrichment**. Under pure opportunity every class should sit at 1.0 per marker. The two long classes read **3.33 and 0.22** — so long-and-dense is enriched 3.3-fold *and* long-and-sparse is depleted 4.5-fold, a 15-fold spread in which the depletion is as large as the enrichment. A mechanism that only enriches dense units would not account for it, which makes the **depleted** arm the more diagnostic half should the residual ever matter.

No account is offered. Four mechanisms have been proposed for a quantity that turned out to be nine tenths arithmetic, and "clusters with more markers contain more QTN, plus a residual spread" is where this closes.

16 The distance threshold: the parameter nobody was checking

Every script in this module hand-sets `distance_threshold = 1e5`. So does the panel session's. The package does **not** require this: with `distance_threshold = NULL` it derives a per-chromosome value, $d = d_{\text{from\_rho}}(a_{\text{pred}}, \rho) = \rho/(a(1-\rho))$, exactly as `min_r2` is already derived. Both sessions had been overriding a rule with a constant, and neither had noticed.

16.1 A withdrawn result: "inert on bgs5"

This section first reported that the substitution is **inert** here, on the grounds that only 0.004% of gaps exceed 1e5 and the derived value exceeds the largest observed gap on 20 of 20 chromosomes. **That was computed on the wrong quantity and the conclusion is withdrawn.** 2c found it by reading the source:

```
split_by_distance <- function(pos_min, pos_max, distance_threshold) {
  gap <- pos_min[ord] - c(NA, pos_max[ord][-length(ord)])
  new_run <- is.na(gap) | gap > distance_threshold
```

The gap is between consecutive **stage-1 clusters**, and it is applied to the **flagged subset only** — clusters with any member $ld_w >$ `ld_w_threshold` (`ld_prune_and_eMLG.R:389`). Both sessions had computed gaps between adjacent *markers* over the whole map. Wrong unit and wrong subset, and the subset dominates: flagged clusters are sparse, so gaps between them are far larger.

	as first reported	corrected
quantity	adjacent markers, whole map	flagged stage-1 clusters
gaps exceeding 1e5	0.004%	**1.54%** pooled, **0.60–10.94%** by setting
derived > largest gap	160 of 160 chr	**85 of 160**
split points lost, 1e5 → derived	(none)	**68–100%** in every setting

So the substitution is **consequential on both datasets**, and the inert-versus-dominant asymmetry both sessions were reporting does not exist. 2c's own figure moves too — their "4 gaps exceed 1e5" becomes 453, and their 1,300-fold becomes 9–23-fold — but their *direction* survives and mine did not, because mine was the claim that nothing happens.

The dcap operating point does not transfer across simulation settings either, and it is not an independent axis. Gaps exceeding 1e5 span 0.60% to 10.94% across the eight settings, an 18× spread, tracking the number of flagged clusters (1,272 to 32,318, a 25× spread) at Spearman = −0.905. Since `ld_w_threshold` is what sets that count, **`ld_w_threshold` and `distance_threshold` are coupled by construction** — a factorial grid cannot vary them orthogonally and read the margins separately.

On the panel the derived threshold is smaller than the hand-set one — 3–18 kb against 100 kb — so it cuts into the body of the gap distribution rather than sitting past the top of it.

That is an LDscnR calibration issue rather than a parameter choice. $d = 19/a$ at $\rho = 0.95$, so the window **collapses as decay speeds up** — and it carries no term for the size of the feature stage-2 merging exists to bridge. Inversions and cold blocks do not shrink because LD decays faster elsewhere in the genome. On the panel the derived rule produces a **6 kb median** merge distance on a genome whose motivating feature is a 430 kb inversion, so the hand-set 1e5 was not an arbitrary constant: it was compensating for a formula that cannot reach the features.

17 The decay window: 20, not 50

Fitting `n_win_decay` at 10 / 20 / 50 with everything else held at the bundles' own settings, RNG pinned in the caller (the installed LDscnR predates `compute_LD_decay`'s `seed` argument):

w	windows	median a	a spread	b	c	disagreement
10	18	1.87e-05	5.48×	0.03048	0.9936	0.273
20	38	2.56e-05	4.35×	0.03048	0.9941	**0.095**
50	96	3.18e-05	2.99×	0.03048	0.9801	0.159

Paired over files, 20 beats 10 on **9 of 10** (sign $p = 0.021$); 50 is 5 of 10 against 20 ($p = 1.0$) for 2.5× the windows.

Two errors on the way, both recorded because they are the kind that survive:

A five-file subset looked monotone through 50 (0.286 → 0.088 → 0.038) and would have supported it. On all ten it is not monotone, and c also degrades at 50 — the wider window changes the character of the fit rather than sharpening it. Note a's *spread* keeps shrinking to 50; that compression is over-smoothing, not sharpening, so spread is not a quality measure.

The metric was narrower than its name. $|a - a_{\text{pred}}|/a_{\text{pred}}$ reads as distance from a size-corrected prediction, but each bundle holds two *equal-size* chromosomes, so a_{pred} is exactly their mean and the deviation is identical for both by arithmetic. It measures between-chromosome disagreement within one simulation. The independent unit is the **file** ($n = 10$), not the chromosome ($n = 20$); the sign tests were first run at $n = 20$ on duplicated values.

Also: b is identical across all three windows, which is the check that pinning the RNG worked. And a **reorders** — Spearman$(a_{w10}, a_{w50}) = 0.60$, median change 72% — so any between-chromosome a ranking is not robust to this setting.

18 The recombination proxy: T1 passes, T2 pre-registered

PK's argument for adapting `distance_threshold` to the local recombination landscape rests on LD decay being a usable proxy for it. The sims are the right instrument: the true rate is a simulation *input*, and the cM map increments at 10,123 distinct positions on a 29.9 Mb chromosome — **roughly 1.1 kb effective resolution**, against the panel's 838 bins at a median of 489 kb.

No new fit was needed. `LD_decay$by_chr[[chr]]$decay` already stores a per-window $a/b/c$ with start/end.

w	med span	Spearman vs **true** cM/Mb	positive	$a = 0$	$a > 0$ only
5	3.77 Mb	0.690	19/20	2.44%	0.690
10	1.89 Mb	0.758	20/20	3.86%	0.758
20	1.01 Mb	0.830	20/20	6.98%	**0.782**
50	0.41 Mb	0.829	20/20	12.60%	0.742

T1 passes: median within-chromosome Spearman +0.757 at the built setting, 20 of 20 positive, sign $p = 1.9 \times 10^{-6}$. An *external* criterion therefore also picks $w = 20$, agreeing with §17's internal one.

The plateau at 20–50 was an artefact. The headline figure is a composite of two capabilities — *detecting* cold blocks ($a = 0$, where the median true rate is exactly 0.000 at $w \geq 20$) and *tracking* the rate continuously elsewhere. Flat from 20 to 50 only because detection improves with resolution while continuous tracking degrades, the two cancelling. On the continuous part there is a clear peak at 20 and a real fall at 50. **The operational number is 0.782, not 0.830**, because the $a = 0$ windows are exactly the ones a cap overrides.

18.1 What bgs5 contains

Nemo has no structural variants, so the analogue of an inversion is a contiguous zero-recombination block. 24.1% of 50 kb bins have exactly zero recombination: **459 blocks, median 100 kb, q90 550 kb, max 3.7 Mb; 55 at or above 430 kb**, 18 over 1 Mb, 103 Mb of 427 Mb inside one. So bgs5 holds 55 analogues of the single feature motivating a local threshold on the panel.

18.2 T2, pre-registered

T1 does not carry T2. Cluster geometry is 30× enriched for containing QTN and no restriction built on it beats using every cluster (§11); a quantity can track truth and still not improve the partition. T2 asks whether $d = f(a_{\text{local}})$ beats the flat threshold on QTN precision **and** recall; T2b runs it inside versus outside cold blocks, where a flat threshold should fail if it fails anywhere. Recorded before running: **if T1 passes and T2 fails, the conclusion is that the proxy is sound and the pipeline should not use it.**

19 Three transfer functions, all refuted the same way

$d = \rho/(a(1-\rho))$ diverges as $a \to 0$, and here $a \to 0$ is a *correct* answer. Three fixes were proposed and all three failed:

Proposal	**Why it failed**
Shrink a_{local} toward the chromosome fit where the window is poorly determined	The degenerate windows are **not** poorly determined. Median true rate 0.083 cM/Mb against 3.310 elsewhere (Wilcoxon $p = 1.1 \times 10^{-9}$) at essentially unchanged fit quality (contrast 0.948 against 0.965) — cold blocks reported correctly. Shrinking erases the signal that motivates locality, and no fit-quality diagnostic separates them.
Cap at feature extent (block q90, 550 kb)	An order of magnitude too tight: costs 0.830 → 0.656 to buy nothing a generous cap does not.
Floor at feature extent	Measured on the panel: binds on **98.4%** of windows, so the rule outputs one constant across the genome. A clamp whose lower bound binds almost everywhere is a flat threshold with extra steps.

All three act hardest exactly where the local estimate is most informative. Two were proposed here and only the third was flagged in advance. The criterion is now stated for any fourth proposal: **check whether it acts most aggressively on the windows that motivate locality, before implementing it.**

19.1 Exclusion is not capping, and overstates by 5×

Both sessions quoted *exclusion* numbers as *cap* numbers for several messages. A cap is $d \leftarrow \min(d, \text{cap})$ with the window **retained**, so it keeps that window's rank against the uncapped ones; removal does not.

cap	250 kb	550 kb	1 Mb	2.1 Mb	3.7 Mb	10 Mb	30 Mb	none
at ceiling	89.9%	65.1%	45.7%	32.3%	26.2%	17.5%	10.8%	0.0%
ρ	0.381	0.656	0.755	0.793	0.803	0.826	0.829	0.830

The cost is **monotone in the cap**, so there is no optimum and "which cap" is the wrong question. A 10 Mb ceiling costs 0.004 while bounding a quantity that otherwise reaches **4,961 Mb on a 30 Mb chromosome**. The cap belongs where the derived distance stops being physically meaningful, not where the features stop.

A baseline trap, caught only by sweeping: capping at 2.1 Mb first appeared to *improve* things (0.782 → 0.793). It does not — 0.782 is the $a > 0$ subset, since $d = \infty$ is not finite and those windows drop out, while the capped series retains them tied at the ceiling. Like-for-like uncapped is 0.830.

20 The minimum-cluster-size floor is inert above 2

The panel session obtains 1 discovery at floor 2 and 14 at floor 8, raising whether the higher floor *detects* more or merely escapes multiplicity — BH over 2,631 units against 790,578 markers being a 300× smaller correction. The sims can measure the recall cost the panel cannot. 80 panels, floor applied **before** BH so the multiplicity genuinely changes:

arm	units tested	discoveries	precision	recall
single-SNP	292,194	74.9	0.187	**0.378**
floor 1	84,869	20.0	0.350	0.229
floor 2	40,590	23.0	0.338	0.245
floor 5	12,723	23.9	**0.393**	0.253
floor 8	6,657	24.3	0.372	0.261

The means suggest a gentle rise. Paired across panels with ties, it is not there: floor 2 → 8 gives recall 35 W / 22 L / 23 T ($p = 0.11$), precision 38 / 30 / 12 ($p = 0.40$), discoveries 43 / 26 / 11 ($p = 0.053$). **Above 2 the floor is inert.**

A predicted asymmetry is therefore **refuted**: the claim was that if the panel gains while the sims *lose*, that is what multiplicity relief predicts. The sims neither gain nor lose, so there is no asymmetry to interpret. What survives is weaker — if the panel's effect is relief it should scale with the multiplicity escaped, and 6× here yields nothing detectable while 300× there yields 14×. Consistent with relief; not proof.

Floor 1 → 2 gains recall and loses precision (16 W / 38 L, $p = 0.0038$): removing singletons lowers the multiplicity, so more clusters clear BH. Floor 2 remains right on the *structural* ground — Simes over a singleton is that marker's own p, so below 2 the cluster test is not a cluster test — but it must not be sold as a precision improvement, because measured it is not one.

And the headline trade-off is unchanged at every floor: single-SNP beats every floor on recall (0 W / 63 L against floor 2) and loses to every floor on precision (60 W / 11 L). The floor moves along the clustering trade-off; it does not change it.

21 What the empirical panel can and cannot do

Two independent findings landed on the same conclusion, so it is stated once:

- **Truth resolution.** The panel's narrowest window (362 kb) is 0.74× its own 489 kb map bins — finer than what it is scored against — so its correlation decline below roughly 500 kb is its map running out, not the proxy degrading. The sims' narrowest is 367× *coarser* than truth.
- **Cap inertness.** Every cap moves the panel's correlation by at most 0.005, because its divergent windows are 0.6–1.8% of the total against a third here.

The panel can falsify the proxy but cannot calibrate it, because calibration needs variation in exactly the regime the panel lacks. This is a property of the *stickleback map and its LD structure*, not of empirical data in general — a species with a dense pedigree map would not have the limit.

The two datasets also differ structurally, confirmed by three independent routes: near-zero a rises with resolution here (2.44 → 12.60%) and *falls* there (1.75 → 0.64%); capping lowers the correlation here and raises it there; and the panel has no zero-recombination bins at all. bgs5 has a cold-block regime and the panel does not. Consequently, should T2 succeed, the honest claim is that a_{local} **helps in the cold-block regime and is inert to degenerate outside it** — not that it helps in general and the panel merely failed to confirm.

22 Provenance

Generation	Maps	Half-decay	Used for
`bgs5`	MAP_RES 3.3e-5; 31.5% of markers share a position	55.8 kb (18 win/chr)	**all cluster-level results and both manhattans**
`pilot_v2_v05c1`	MAP_RES 3.3e-6; no shared positions	13.3 kb (95 win/chr)	marker-level power only; `V0.5_c1` only
`analysis_inputs`	pre-bgs5	not compared	null behaviour only; `V2_c1`, nobgs, `gc_mode = none`

At $\rho = 0.5$ the half-decay **is** the stage-2 clustering distance, so the units themselves would differ on the newer maps. The difference traces to decay fitting scale, not the maps: same-position marker pairs show r^2 of 0.023 against 0.022 for pairs 1 bp–10 kb apart.

Scripts: `filter_then_test_clusters.R`, `filter_then_test.R`, `filter_fdr_check.R`, `stratified_paired_table.R`, `recluster_filter_test.R`, and the three `plot_*` scripts. For §16–§21: `operating_points.R`, `decay_window_test.R`, `proxy_T1_local_decay.R`, `proxy_T1_by_window.R`, `cap_cost.R`, `floor_sweep.R`, `dcap_operating_point.R`. Parameters: α 0.05; stage-2 `min_r2_rho` 0.5, `distance_threshold` 1e5; scoring `dmax_cap` 1e5. Commits `60c8134`, `88797ff`, `7e595b5`.

Every comparison is **paired within panel**, reported with wins, losses, ties, a median with IQR, and a sign-test p-value. Differences of means are not used.